

Simulation-Driven Optimization of Dual-Coupled Micro-Ring Entanglement Sources for Quantum Optical Networks

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Abstract—Dual-coupled micro-ring (DCM) resonators are a promising on-chip source of frequency-bin entangled photons for quantum optical networks, but their spectral purity is highly sensitive to fabrication tolerances and thermal drift. We present a coupled-mode-theory-based digital twin of the DCM that maps device-level parameters – inter-ring coupling, dispersion, and auxiliary-ring thermal detuning – to network-facing figures of merit including heralded photon purity and Schmidt number. In representative C-band operating conditions, optimal thermal tuning of the auxiliary cavity raises heralded purity by up to 75% at fixed brightness and reduces the Schmidt number from approximately 1.8 to 1.2, enabling predictable, pre-deployment configuration of integrated entanglement sources in hybrid quantum-classical WDM networks.

Index Terms—Quantum optical networks, integrated photonics, frequency-bin entanglement, dual-coupled micro-ring resonators, digital twin, quantum key distribution.

I. INTRODUCTION

Recent advances in photonic integrated circuits (PICs) have enabled the generation and manipulation of entangled photons directly on-chip [1]–[4], setting the stage for a new generation of scalable quantum networks [5]–[7]. Microresonator-based sources in particular combine high quality factors and small footprints with the ability to produce broadband biphoton combs [8], and frequency-bin entanglement is especially well suited for integrated photonics because it leverages the spectral multiplexing and stability inherent to resonator-based systems [9]. Existing single micro-ring sources, however, remain highly sensitive to small geometric variations, uncontrolled dispersion, and thermal drift, which limit their reliability when integrated into deployed optical fiber networks.

To mitigate these issues, we adopt a dual-coupled micro-ring (DCM) architecture for our entanglement sources. The addition of a thermally tunable auxiliary ring introduces a controllable degree of freedom over the dispersion profile [10], [11], yielding significant improvements in the spectral purity and stability of the generated pairs – a key requirement for entanglement-based QKD [12]. Operating in the C-band

(1550 nm) with 50 GHz channel spacing, the DCM is directly compatible with existing telecom hardware: quantum and classical channels can co-propagate over the same shared fiber infrastructure, as illustrated in Fig. 1a [13], [14].

In classical optical networking, deploying a new physical-layer component typically relies on predictive models that connect device-level parameters to system-level performance metrics [15], [16]. For quantum optical networks, such model-based methodologies remain largely underdeveloped, particularly for integrated entangled-photon sources whose performance depends sensitively on fabrication and environmental variations and is therefore difficult to incorporate into existing network planning and control frameworks.

In this work we address this gap by introducing a digital twin of the DCM entanglement source. The twin is based on a coupled-mode-theory (CMT) model and explicitly maps device-level parameters – inter-ring coupling, dispersion, and auxiliary-ring thermal detuning – to network-facing figures of merit including heralded photon purity and Schmidt number. The contribution of this paper is therefore not a single optimized operating point, but the predictive framework itself, which can be deployed alongside a fabricated device to support pre-deployment optimization, yield improvement, and remote, network-aware reconfiguration. In representative C-band operating conditions, the twin predicts purity enhancements exceeding 70% at fixed brightness and a corresponding reduction of the effective Schmidt number, directly translating into improved system-level performance for entanglement-based quantum communication.

II. BACKGROUND AND MOTIVATION

For quantum networks to be adopted at scale, their quantum components must be compatible and integrable with existing communication infrastructures. Entangled photon sources act as critical nodes in such networks, supplying photon pairs correlated in polarization, time, or frequency. Previous demonstrations [17] have shown how single micro-ring resonators (Fig. 1b) can generate bi-photon frequency combs, but such

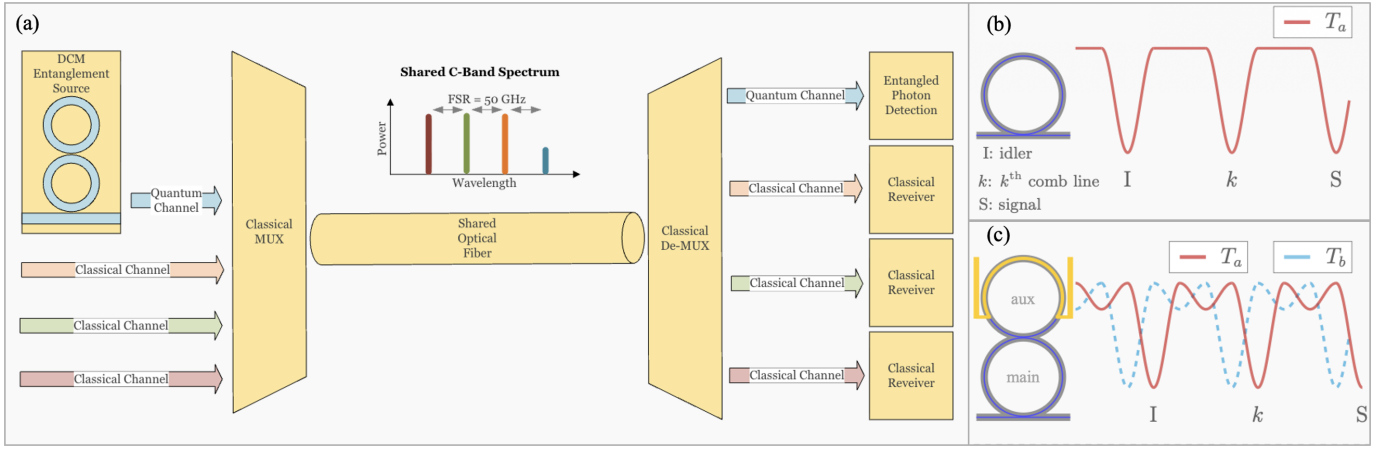


Fig. 1. (a) Integration of the DCM source into a hybrid quantum-classical WDM fiber network, where quantum and classical channels co-propagate over shared fiber infrastructure and are demultiplexed at the network edge. (b) Single microring resonator producing signal (S) and idler (I) modes with fixed transmission T_a . (c) Dual-coupled microring with tunable auxiliary ring showing modified resonances T_a and T_b .

single-ring designs offer limited control over dispersion and spectral coherence. The DCMs developed by our group [18] address this limitation by introducing a tunable auxiliary cavity for spectral engineering.

The DCM response, however, depends on many interacting variables – thermal detuning, inter-ring coupling, dispersion, and pump conditions. Experimentally identifying an optimal operating point is therefore time-consuming and sensitive to fabrication variability. The challenge becomes more acute in networked scenarios, where sources must be configured remotely and operate reliably over long time scales. A predictive model that can be queried before and during deployment is essential to pre-determine tuning parameters, improve design yield, and accelerate experimental verification.

III. DCM WORKING PRINCIPLES

From a network-engineering perspective, the appeal of the DCM architecture lies not only in its spectral flexibility, but also in its ability to deterministically map control parameters, such as thermal detuning, to changes in network-facing performance metrics.

The DCM structure (Fig. 1c) comprises two adjacent micro-rings: a main ring, in which photon pairs are generated via spontaneous four-wave mixing (SFWM), and an auxiliary ring, whose refractive index can be controllably modified through local thermal tuning. Heating the auxiliary ring shifts its effective refractive index, thereby reshaping the coupled-mode response of the DCM and altering both the density and location of the spectral lines of the generated photon pairs. This additional degree of freedom can be used to suppress unwanted mode interactions, align the resonator response with a phase-matched operating point, and improve photon purity and entanglement stability. The DCM thus combines the advantages of single micro-rings – high Q factor and small footprint – with fine spectral tuning, providing a compact platform for bridging on-chip quantum sources and installed fiber optical communication systems.

IV. SIMULATION ENVIRONMENT

The simulation environment functions as a digital twin of the DCM and is based on coupled-mode theory (CMT) [19], applied here to both a single micro-ring resonator and the DCM [10], [20]:

$$\frac{da_1}{dt} = \left(i\Delta_1 - \frac{\kappa_1}{2}\right) a_1 + iJ a_2 + \sqrt{\kappa_{\text{ex}}} s_{\text{in}}, \quad (1a)$$

$$\frac{da_2}{dt} = \left(i\Delta_2 - \frac{\kappa_2}{2}\right) a_2 + iJ a_1. \quad (1b)$$

where a_1 and a_2 are intracavity fields, Δ_i detunings, κ_i total decay rates, κ_{ex} external coupling, and J the inter-ring coupling coefficient. In the single-ring case, the inter-ring term ($iJ a_2$) is absent, yielding a standard Lorentzian resonance.

From Eqs. (1), the cavity transfer functions $H_{1,2}(\omega)$ are derived and used to compute the **Joint Spectral Amplitude (JSA)** of the photon pairs as

$$f(\omega_s, \omega_i) \propto \alpha(\omega_s + \omega_i) H_1(\omega_s) H_1(\omega_i) \Phi(\omega_s, \omega_i), \quad (2)$$

where $\alpha(\omega)$ is the pump spectral envelope, $H_1(\omega)$ represents the frequency-domain transfer function of the primary ring obtained from the CMT model, and $\Phi(\omega_s, \omega_i)$ is the dispersion-dependent phase-matching term. The corresponding **Joint Spectral Intensity (JSI)** (Fig. 2a) is $|f(\omega_s, \omega_i)|^2$.

The model computes the optical response in the frequency domain and predicts the SFWM pair-generation rate as a function of quality factor Q , dispersion (β_2), and coupling coefficient κ [21]. From the JSI we extract the **Schmidt number** K and the **heralded photon purity** $P = 1/K$ as indicators of spectral entanglement, which in turn upper-bound the achievable two-photon interference visibility and therefore the secure key rate (SKR) of any entanglement-based QKD protocol built on top of the source.

The Green's-function analysis (Fig. 2b) shows how thermal tuning of the auxiliary ring reshapes the drop-port resonance profile, while the brightness-purity trade-off (Fig. 2c)

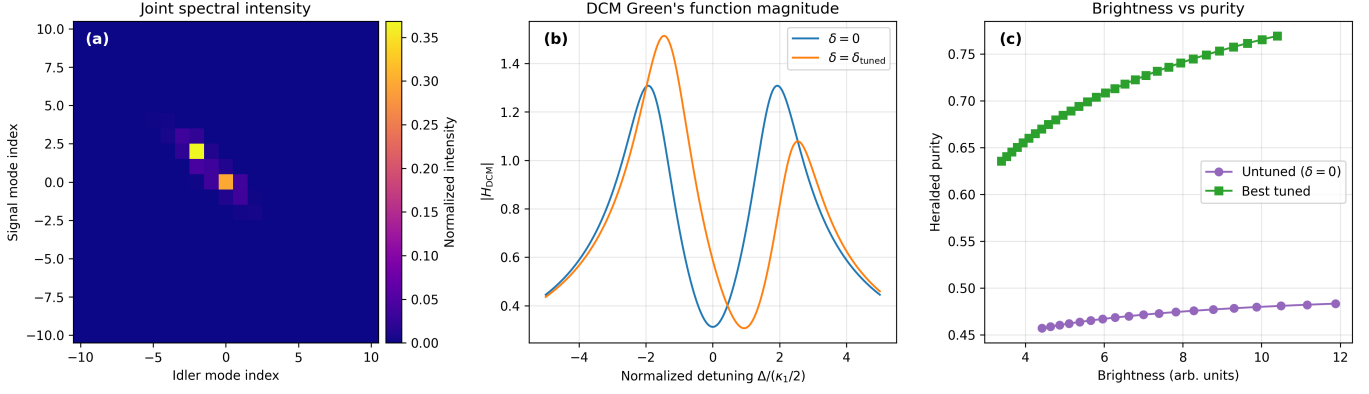


Fig. 2. Simulation outputs illustrating the impact of thermal tuning on network-relevant source characteristics. (a) Joint spectral intensity (JSI) of the generated photon pairs, showing frequency anti-correlation between signal and idler modes. (b) Normalized magnitude of the DCM Green's function $|H_{DCM}|$ (drop-port response) for the untuned ($\delta = 0$) and thermally tuned ($\delta = \delta_{\text{tuned}}$) cases. (c) Brightness–purity trade-off for untuned and optimally tuned DCM operation, computed at equal pump power and coupling conditions.

TABLE I
REPRESENTATIVE DEVICE AND NETWORK PARAMETERS USED IN THE SIMULATION.

Parameter	Symbol	Value
Operating band	–	C-band, 1550 nm
Channel spacing (FSR)	$\Delta\nu$	50 GHz
Intrinsic quality factor	Q_i	$\sim 10^6$
Main-ring decay rate	κ_1	$2\pi \times 0.2$ GHz
Auxiliary-ring decay rate	κ_2	$2\pi \times 0.2$ GHz
Inter-ring coupling	J	$2\pi \times 0.4$ GHz
Group-velocity dispersion	β_2	-100 ps ² /km
Pump power (on-chip)	P_p	1 mW

demonstrates that an optimally tuned DCM achieves higher purity at equal or lower brightness than the untuned case. Representative device and network parameters used throughout the simulation are summarized in Table I.

V. NETWORK-LEVEL IMPLICATIONS

The purity and Schmidt-number predictions of the simulation map directly onto the performance of entanglement-based QKD protocols deployed over shared optical infrastructure. In the BBM92 protocol, the two-photon interference visibility V achievable at the receiver upper-bounds the asymptotic QBER through $\text{QBER} \geq (1 - V)/2$, with V itself bounded by the spectral purity P of the source. A 75% increase in heralded purity therefore directly suppresses the source-limited contribution to QBER, relaxing the noise budget available to Raman scattering and dark counts in a coexistence scenario with classical WDM traffic [13], [14]. Equivalently, at a fixed target QBER, this translates into an extension of the secure-fiber reach of the link.

Beyond a single link, the key property of a predictive device model is that the mapping from on-chip heater currents to network-facing figures of merit can be queried in closed form. This interface is precisely what is needed to integrate quantum

sources into the model-based planning and control frameworks already in use for transparent optical networks [16], [22], opening the door to quantum-aware routing and wavelength assignment across shared WDM grids where entangled-photon and classical channels must coexist.

VI. CONCLUSIONS

We have presented a digital twin of the dual-coupled micro-ring entanglement source, built on a coupled-mode-theory model and validated against the design space of our fabricated DCM devices. In representative operating conditions, tuning the auxiliary-cavity detuning raises the heralded purity by up to 75% at fixed brightness, lowers the Schmidt number from approximately 1.8 to 1.2, and largely suppresses unwanted spectral correlations – corresponding to an estimated 30% improvement in the brightness–purity trade-off that is directly relevant to QKD performance.

More broadly, these results show how a predictive model-based workflow can transform integrated entanglement sources from finely tuned laboratory devices into predictable, network-compatible components. By acting as a digital twin, the proposed framework enables pre-deployment optimization, reduces experimental calibration effort, and supports adaptive configuration aligned with network-level performance objectives. Future work will close the validation loop against measured DCM transmission spectra and extend the framework to multi-source network topologies, where several DCM devices must be co-configured across WDM channels – a practical step toward scalable quantum–classical optical networks and future quantum internet architectures.

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